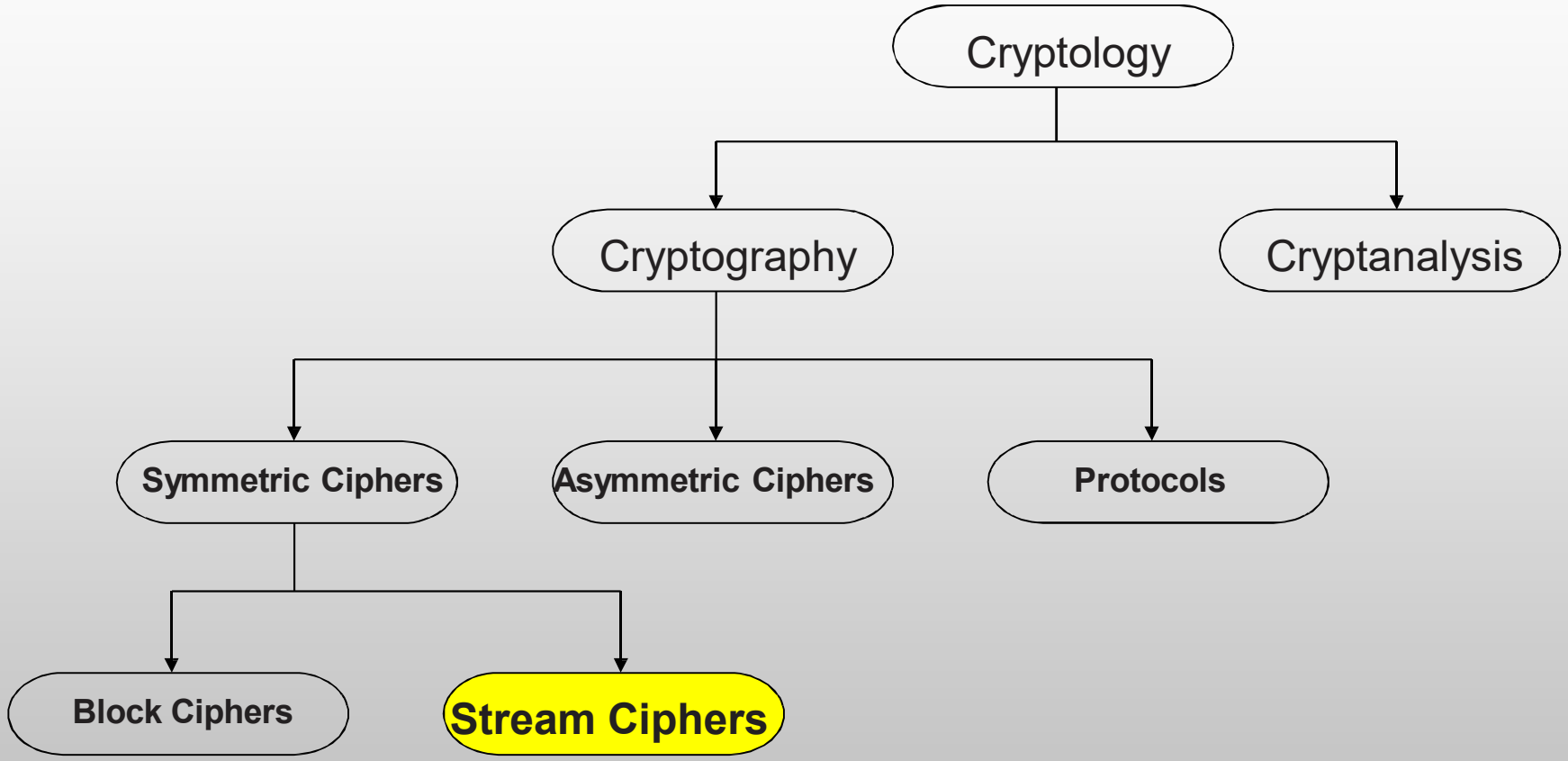
Stream Ciphers in the Field of Cryptology

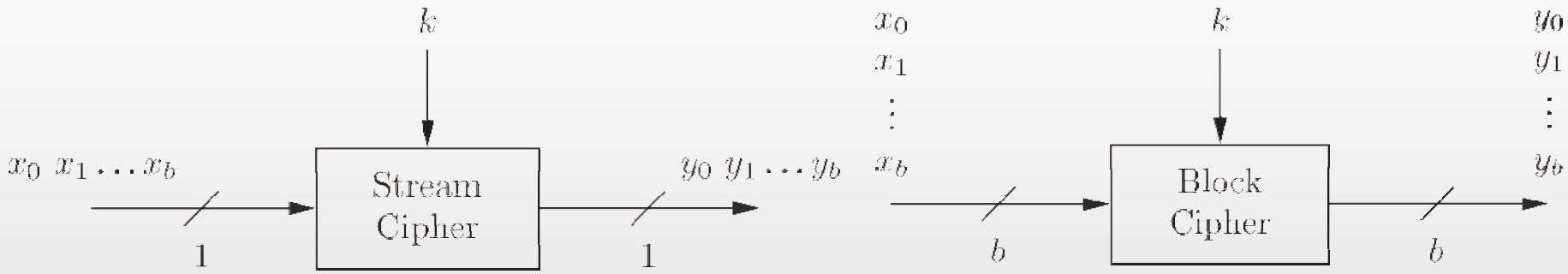


Stream Ciphers were invented in 1917 by Gilbert Vernam

STREAM CIPHERS



Stream Cipher vs. Block Cipher

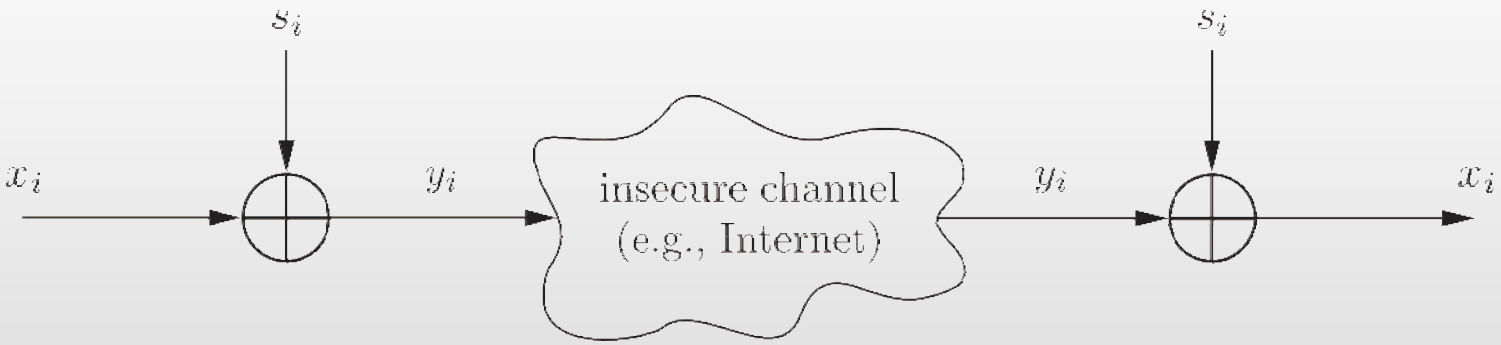


- **Stream Ciphers**
  - Encrypt bits individually
  - Usually small and fast common in embedded devices (e.g., A5/1 for GSM phones)

- **Block Ciphers:**
  - Always encrypt a full block (several bits)
  - Are common for Internet applications

Encryption and Decryption with Stream Ciphers

Plaintext x_i, ciphertext y_i and key stream s_i consist of individual bits



- Encryption and decryption are simple additions modulo 2 (aka XOR)
- Encryption and decryption are the same functions

- **Encryption:** $y_i = e_{si}(x_i) = x_i + s_i \bmod 2$ $\quad x_i, y_i, s_i \in \{0,1\}$
- **Decryption:** $x_i = e_{si}(y_i) = y_i + s_i \bmod 2$

Why is Modulo 2 Addition a Good Encryption Function?

- Modulo 2 addition is equivalent to XOR operation
- For perfectly random key stream s_i , each ciphertext output bit has a 50% chance to be 0 or 1

  Good statistic property for ciphertext
- Inverting XOR is simple, since it is the same XOR operation

x_i	s_i	y_i
0	0	0
0	1	1
1	0	1
1	1	0

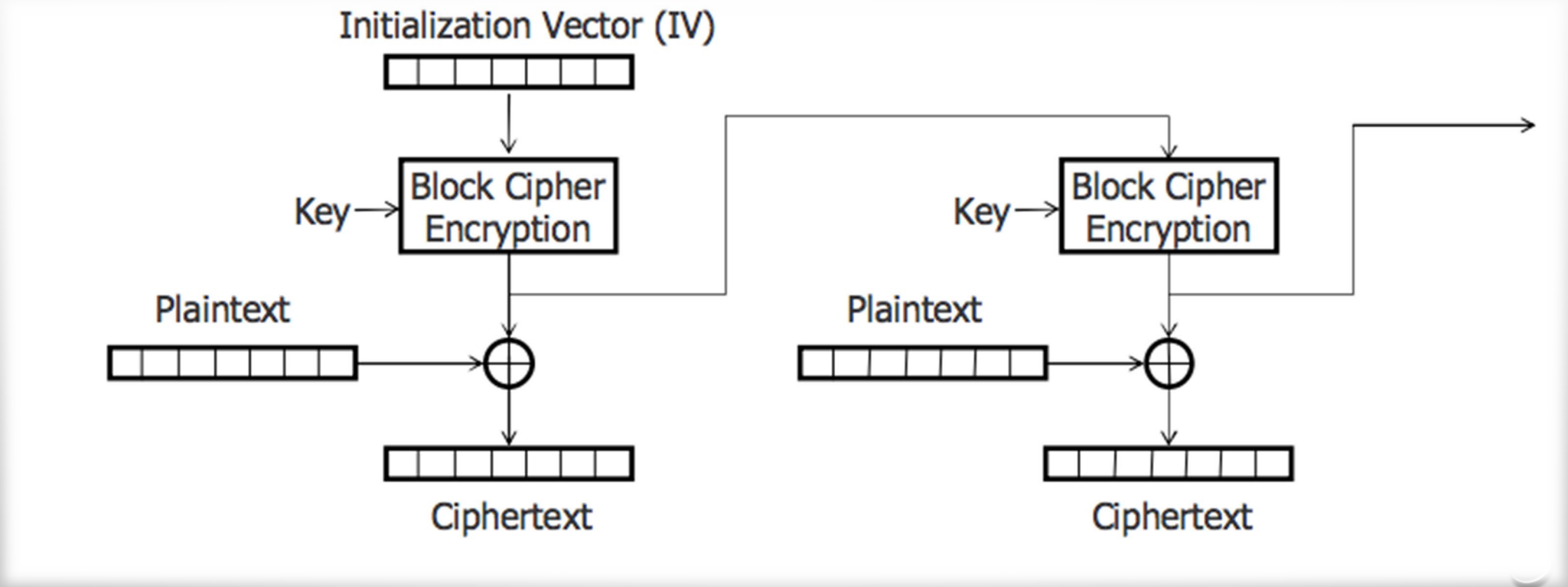
Stream Cipher: Throughput

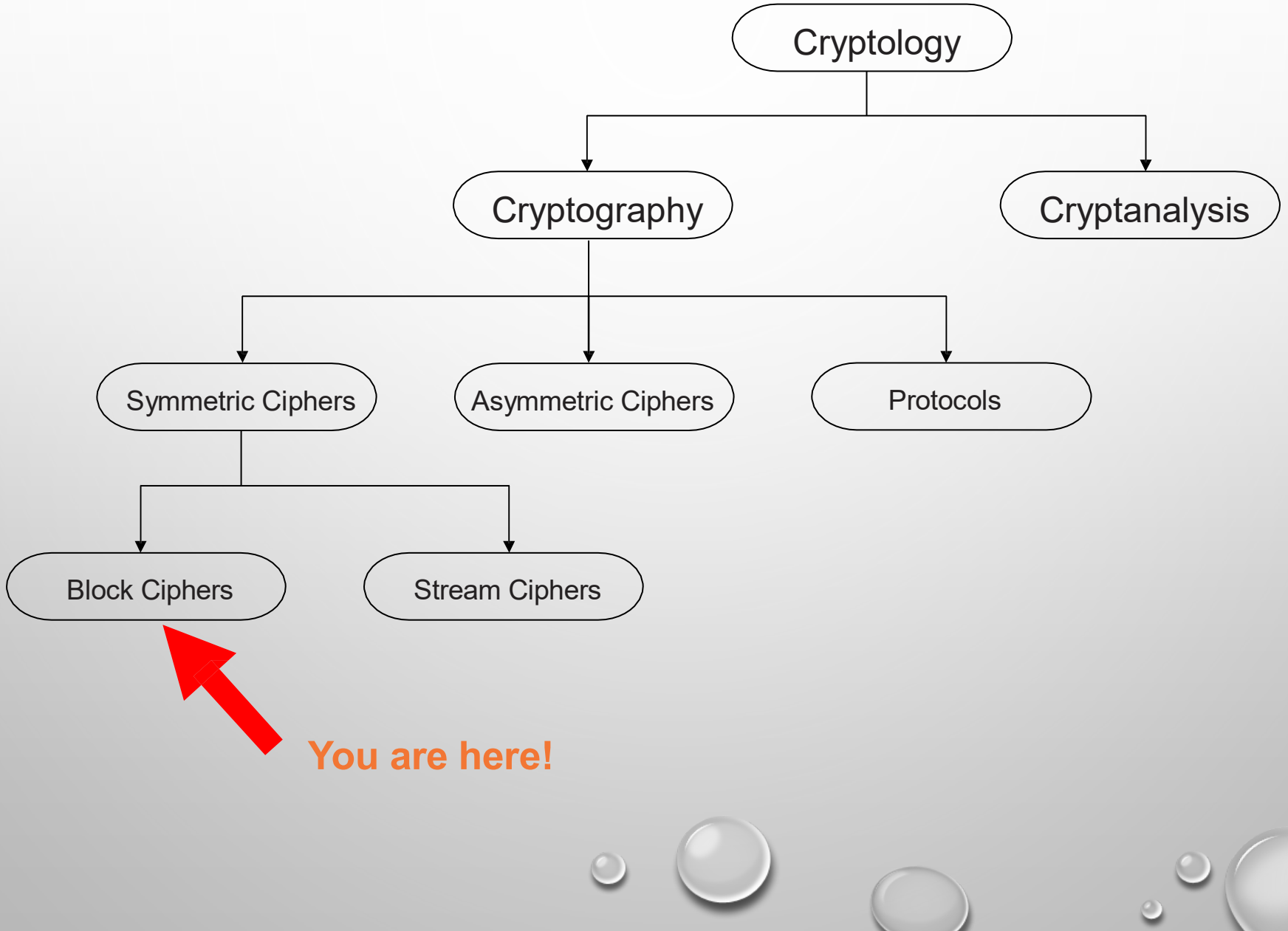
Performance comparison of symmetric ciphers (Pentium4):

Cipher	Key length	Mbit/s
DES	56	36.95
3DES	112	13.32
AES	128	51.19
RC4 (stream cipher)	(choosable)	211.34

Source: Zhao et al., Anatomy and Performance of SSL Processing, ISPASS 2005

BLOCK CIPHERS




Cryptology
Cryptography
Cryptanalysis
Symmetric Ciphers
Asymmetric Ciphers
Protocols
Block Ciphers
Stream Ciphers
You are here!

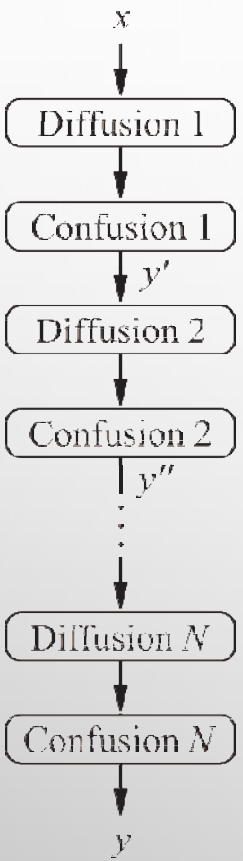
Block Cipher Primitives: Confusion and Diffusion

- Claude Shannon: There are two primitive operations with which strong encryption algorithms can be built:
  1. **Confusion:** An encryption operation where the **relationship between key and ciphertext is obscured**.
     Today, a common element for achieving confusion is **substitution**, which is found in both AES and DES.

  2. **Diffusion:** An encryption operation where the **influence of one plaintext symbol is spread over many ciphertext symbols** with the goal of hiding statistical properties of the plaintext.
     A simple diffusion element is the **bit permutation**, which is frequently used within DES.

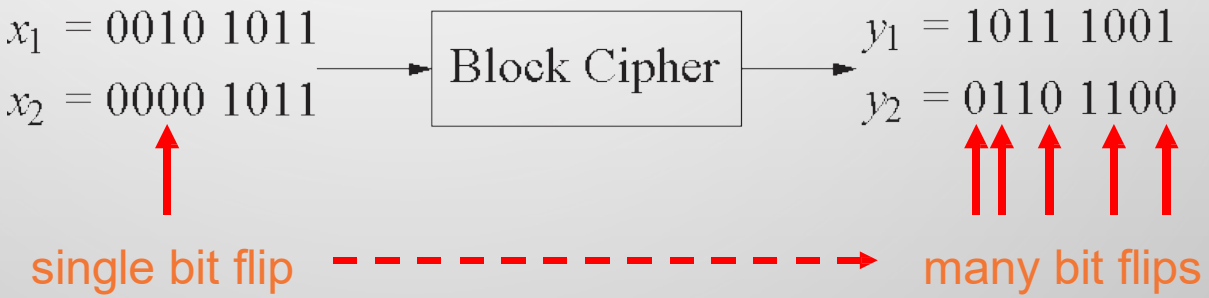
- Both operations by themselves cannot provide security. The idea is to concatenate confusion and diffusion elements to build so called *product ciphers.*

Product Ciphers



- Most of today's block ciphers are *product ciphers* as they consist of rounds which are applied repeatedly to the data.
- Can reach excellent diffusion: **changing of one bit of plaintext results** *on average* in the **change of half the output bits**.

Example:



HOW TO CHOOSE AN ENCRYPTION ALGORITHM?

	DES	3DES	AES
Is the algorithm trusted by the cryptographic community?	Been replaced by 3DES	Yes	Verdict is still out
Does the algorithm adequately protect against brute-force attacks?	No	Yes	Yes

DATA ENCRYPTION STANDARD (DES)

- The most popular symmetric encryption standards.
  - Developed by IBM
  - Thought to be unbreakable in the 1970s
  - Shared keys enable the encryption and decryption

- DES converts blocks of 64-bits of clear text into ciphertext by using an encryption algorithm.
  - The decryption algorithm on the remote end restores ciphertext to clear text.

DES SCORECARD

Description	Data Encryption Standard
Timeline	Standardized 1976
Type of Algorithm	Symmetric
Key size (in bits)	56 bits
Speed	Medium
Time to crack	The EFF's DES cracker (Deep Crack) breaks a DES key in 56 hours
Resource Consumption	Medium

DES SECURITY RATING

- Because of its short key length, DES is considered a good protocol to protect data for a very short time.
  - **3DES** is a better choice to protect data because it has an algorithm that is very trusted and has higher security strength.
- Recommendations:
  - Change keys frequently to help prevent brute-force attacks.
  - Use a secure channel to communicate the DES key from the sender to the receiver.

Alternatives to DES

Algorithm	I/O Bit	key lengths	remarks
AES / Rijndael	128	128/192/256	DES "replacement", worldwide used standard
Triple DES	64	112 (effective)	conservative choice
Mars	128	128/192/256	AES finalist
RC6	128	128/192/256	AES finalist
Serpent	128	128/192/256	AES finalist
Twofish	128	128/192/256	AES finalist
IDEA	64	128	patented

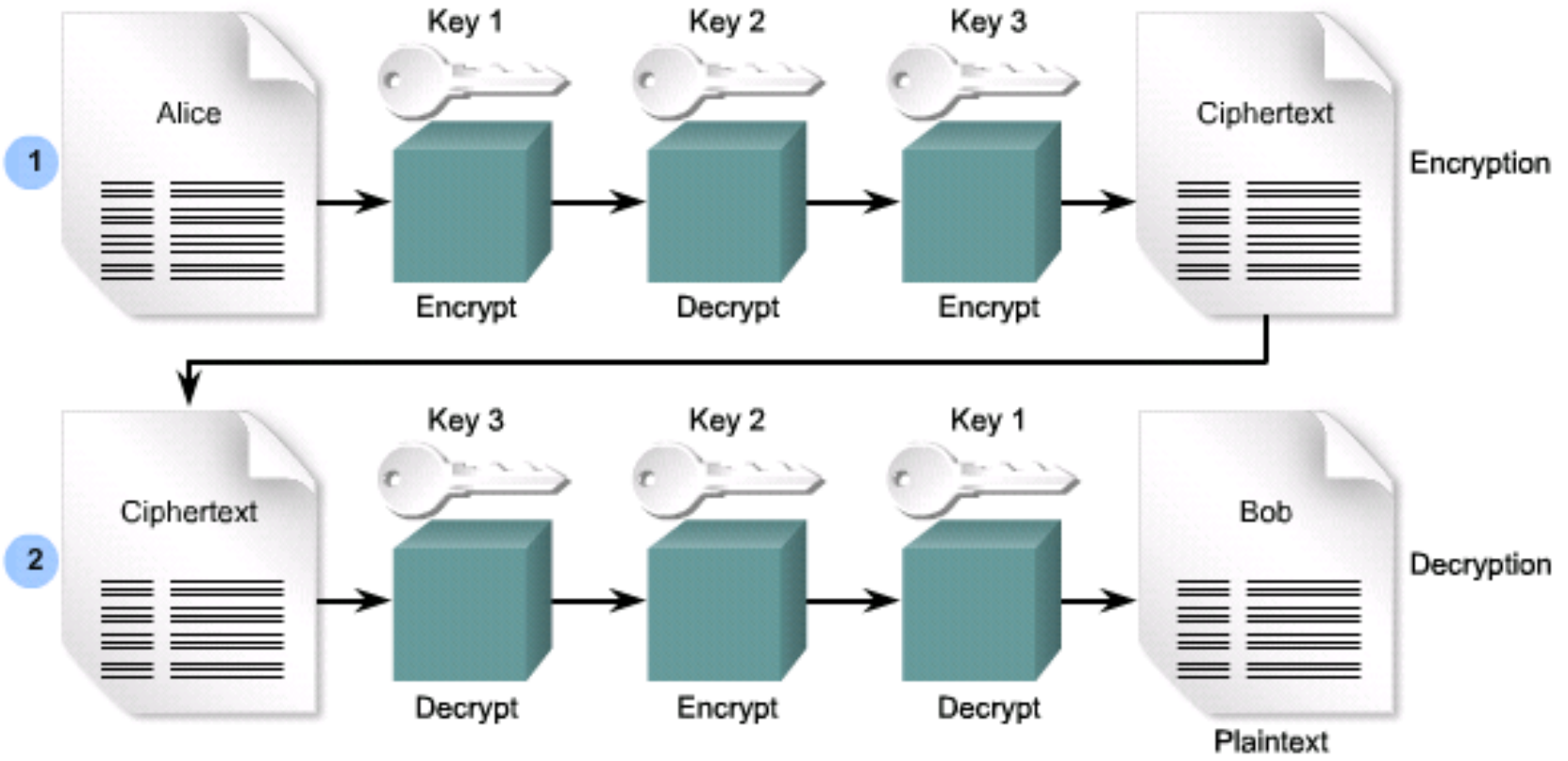
TRIPLE DES (3DES OR TDES)

- 2DES is not recommended. Find out why?
- 3DES is 256 times stronger than DES.
- It takes a 64-bit block of data and performs three des operations in sequence:
  - Encrypts, decrypts, and encrypts.
  - Requires additional processing time.
  - Can use 1, 2, or 3 different keys (when used with only one key, it is the same as des).

3DES SCORECARD

Description	Triple Data Encryption Standard
Timeline	Standardized 1977
Type of Algorithm	Symmetric
Key size (in bits)	112 and 168 bits
Speed	Low
Time to crack (Assuming a computer could try 255 keys per second)	4.6 Billion years with current technology
Resource Consumption	Medium

Symmetric Key (triple DES) Encryption



1. The clear text from Alice is encrypted using Key 1. That ciphertext is decrypted using a different key, Key 2. Finally that ciphertext is encrypted using another key, Key 3.

2. When the 3DES ciphered text is received, the process is reversed. That is, the ciphered text must first be decrypted using Key 3, encrypted using Key 2, and finally decrypted using Key 1.

ADVANCED ENCRYPTION STANDARD (AES)

- AES is an extremely secure Federal Information Processing Standard (FIPS)-approved cryptographic algorithm.
  - Based on the RIJNDAEL algorithm.
  - It use keys with a length of 128, 192, or 256 bits to encrypt blocks with a length of 128, 192, or 256 bits.
    - All 9 combinations of key length and block length are possible.
- AES is the world standard for block encryption.

AES SCORECARD

Description	Advanced Encryption Standard
Timeline	Official Standard since 2001
Type of Algorithm	Symmetric
Key size (in bits)	128, 192, and 256
Speed	High
Time to crack (Assuming a computer could try 255 keys per second)	149 Trillion years
Resource Consumption	Low

RC ALGORITHMS

- The RC algorithms were designed all or in part by Ronald Rivest, who also invented MD5.
- The RC algorithms are widely deployed in many networking applications because of their favorable speed and variable key-length capabilities.
- There are several variation of RC algorithms including:
  - RC2
  - RC4
  - RC5
  - RC6

RON'S CODE OR RIVEST CODES SCORECARD

Description	RC2	RC4	RC5	RC6
Timeline	1987	1987	1994	1998
Type of Algorithm	Block cipher	Stream cipher	Block cipher	Block cipher
Key size (in bits)	40 and 64	1 - 256	0 to 2040 bits (128 suggested)	128, 192, or 256
Use	Variable key-size block cipher that was designed as a "drop-in" replacement for DES.	Most widely used stream cipher based on a variable key-size Vernam stream cipher. It is often used in file encryption products and secure communications, such as within SSL. The cipher can be expected to run very quickly in software and is considered secure.	A fast block cipher that has a variable block size and key size. It can be used as a drop-in replacement for DES if the block size is set to 64-bit.	An AES finalist (Rijndael won). A 128-bit to 256- bit block cipher that was designed by Rivest, Sidney, and Yin and is based on RC5. Its main design goal was to meet the requirement of AES.